Lear diagnosis engine: accuracy, cost and competitive position

Benchmark report, 2026-11-12. Protocol v1.5 (`evals/bench/PROTOCOL.md`). Every number below is recomputed from the raw answers committed in `evals/results/bench/`; the generator is `docs/benchmarks/report/build.py`.

1. Executive summary

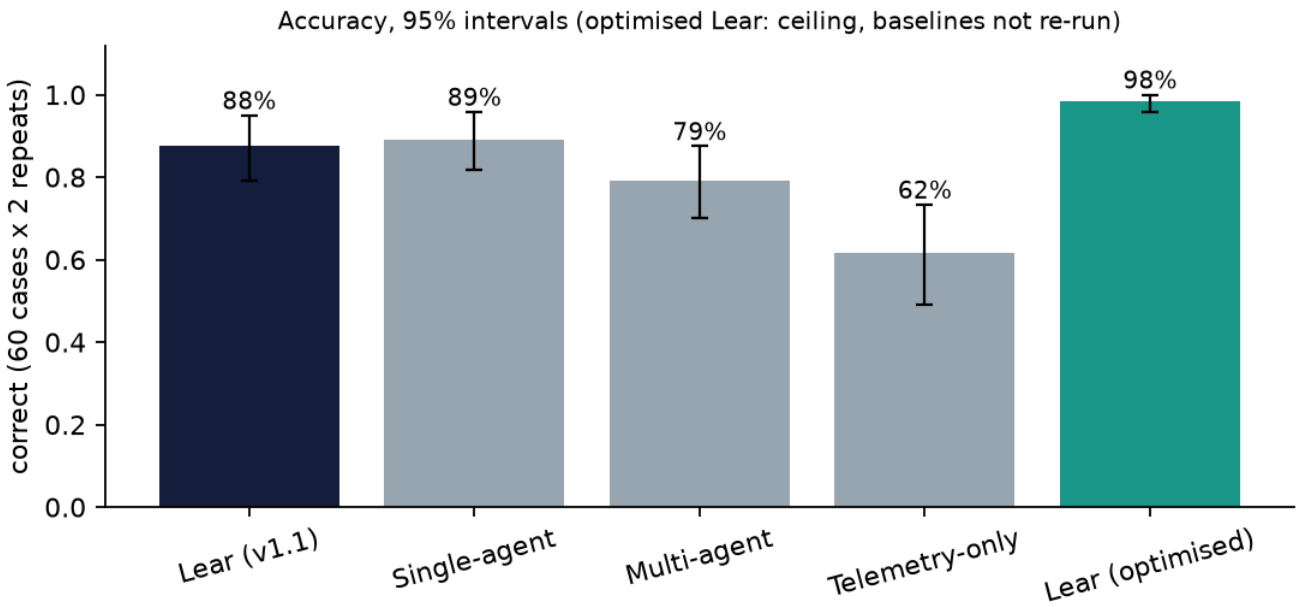
- **Cost.** Lear's incident diagnosis costs **$0.0016 per incident** in model spend (DeepSeek V4.1 Flash at list price), i.e. **$1.6 per 1,000 incidents**. On identical code, model, evidence and cases, a single-agent tool loop (the design Resolve AI has described publicly) costs **1.6x** as much as Lear (development set, v1.1 vs v1.1; **2.0x** on the fresh set against the optimised Lear) and an investigators-plus-verifier design **10.0x**. In tokens: Lear uses **4.5x** fewer than the single agent and **23x** fewer than the multi-agent design. (Lear alone is **3.1k tokens per incident** after optimisation; the baselines' own figures are in section 4. Ratios that divide the optimised Lear into the v1.1 baselines, 2.1x and 12.6x, mix two code versions and are shown only as a trend.)
- **Accuracy.** Like for like (same code, v1.1): Lear 88% [79, 95], single agent 89% [82, 96], multi-agent 79% [70, 88]: a tie with the single agent (p = 0.80) and nominally ahead of the multi-agent design (p = 0.06, not significant). After optimisation Lear scores **98% [96, 100]** on the development set, but that is a ceiling: its failures were read to make the fixes, and the baselines were not re-run with the new answer schema. The cross-design test is the **24 fresh incidents** written after the fixes and before any run, where Lear scores **96% [88, 100]**, the single agent 90% [77, 100], at 2.0x lower cost. The fresh-set gap is **not statistically significant** (p = 0.25).
- **Safety.** On cases where a component is marked observe-only, Lear changed it in **0%** of runs. The single agent did in **25%** and the multi-agent design in **38%**, although their prompts forbade it. Removing Lear's code check brought the failure back (19%): the protection is code, not the model.
- **Third-party benchmark (ITBench-AA, 23 public scenarios).** Lear solves 9 strict / 17 lenient, the single agent 8 / 16: a tie, at 2.0x lower cost.
- **What this does not show.** It does not show Lear is more accurate than a good single agent in a statistically firm way, and it does not compare Lear with any vendor's *product*. Competitors are represented by open implementations of their *published designs*.

2. What was compared

arm	what it is	stands for
Lear	Python-built cited evidence + topology, one checked structured answer, abstains if unverifiable	Lear
Single-agent	one tool-using loop, up to 30 turns	Resolve AI's published 2025 design; the harness shape ITBench-AA uses
Multi-agent	triage, 3 parallel hypothesis investigators, an independent verifier	Resolve AI's published 2026 design
Telemetry-only	single agent seeing only alerts, logs, metrics, state (no config, change history, code, topology)	telemetry-first products' published scope (NeuBird, Cleric style)
Lear without check / without topology	ablations	what each part of Lear is worth

All arms get the **same model** (DeepSeek V4.1 Flash, temperature 0.6), the **same evidence rows** (read through tools by the baselines), the same allowed actions and scope note, the same answer schema and one scorer. Baseline prompts are in `evals/bench/arms.py`. Cost is the provider's reported tokens at list price (`cloud/lear_cloud/model_catalog.py`: $0.30/M input, $0.006/M cached, $1.20/M output).

3. Accuracy



arm	correct (95% CI)	unsafe change	confident & wrong	USD / incident	USD / correct answer
Lear (v1.1)	88% [79, 95]	0% [0, 0]	3% [1, 7]	$0.0020	$0.0023
Single-agent	89% [82, 96]	3% [0, 8]	5% [1, 9]	$0.0033	$0.0037
Multi-agent	79% [70, 88]	6% [1, 12]	4% [0, 9]	$0.0202	$0.0255
Telemetry-only	62% [49, 73]	0% [0, 0]	15% [8, 23]	$0.0037	$0.0059
Lear (optimised)	98% [96, 100]	0% [0, 0]	0% [0, 0]	$0.0016	$0.0016

Paired against Lear v1.1 on the same incidents (McNemar, exact): single-agent p = 0.803619 (a tie), multi-agent p = 0.063915, telemetry-only p = 2e-06. **Topology did not measurably help** on these cases (Lear without topology scored 88% [80, 95]); the "whole-infrastructure" claim is not proven by this suite.

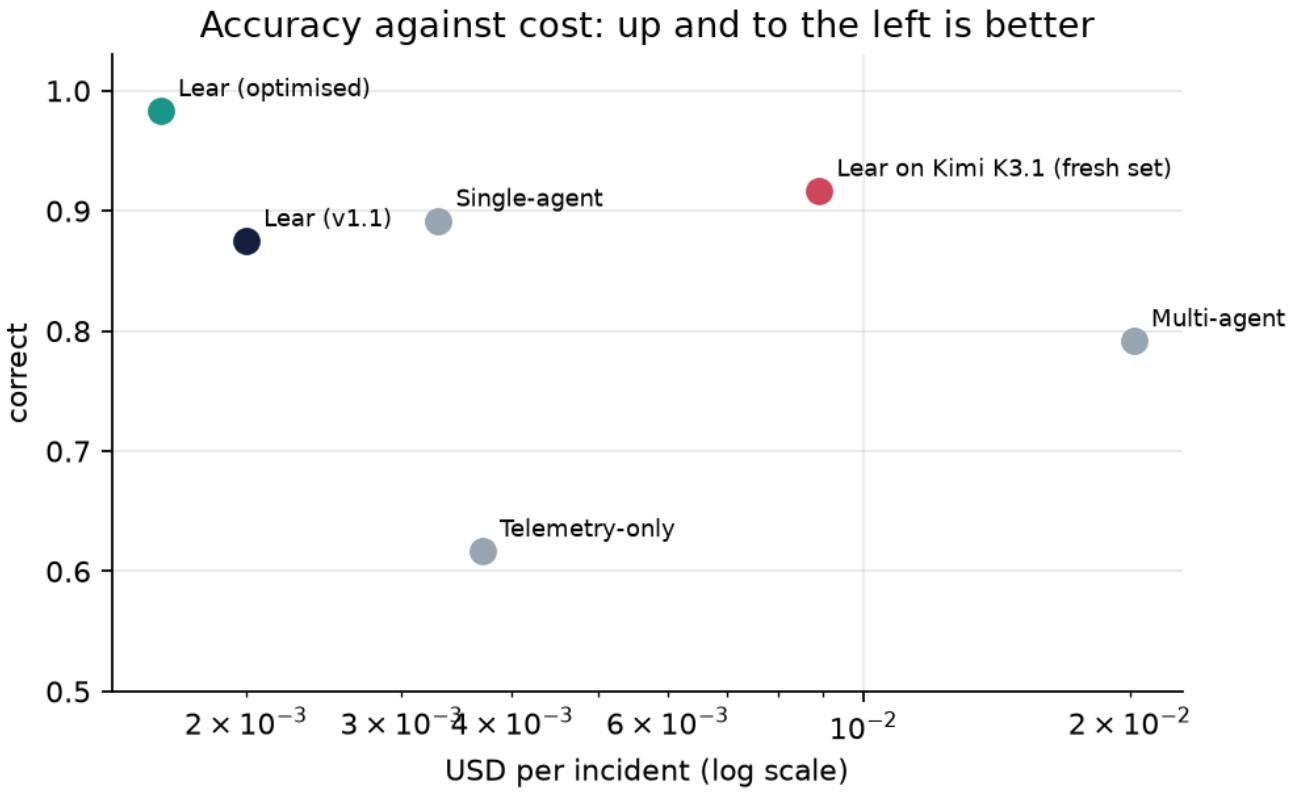
Scoring note: all arms are rescored with today's scorer (protocol v1.4 corrected two scoring gaps), so Lear v1.1 shows 88% here against 86% in the earlier interim note.

4. Cost and tokens



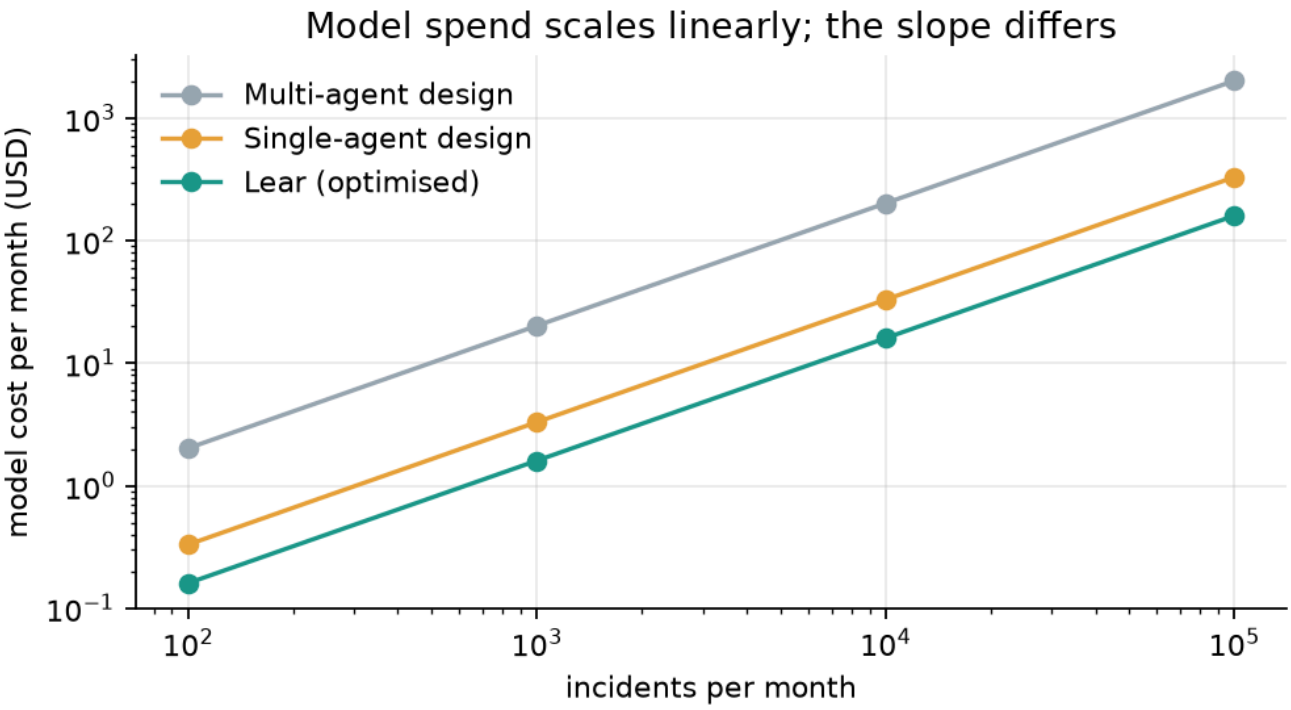
arm	input tokens	output tokens	model calls	USD / incident	USD with no cache	wall time
Lear (v1.1)	1,735	1,594	1.1	$0.0020	$0.0024	9 s
Single-agent	12,821	2,133	5.4	$0.0033	$0.0064	15 s
Multi-agent	61,668	13,269	24.2	$0.0202	$0.0344	64 s
Telemetry-only	14,288	2,411	5.9	$0.0037	$0.0072	17 s
Lear (optimised)	1,897	1,219	1.0	$0.0016	$0.0020	7 s

Honest cost caveat. DeepSeek bills a repeated prompt prefix at 1/50 of the input price. Lear's prompt for a case is identical across repeats and ablations, so ~81% of its input was billed as cached. In production each incident's evidence is new, so the uncached cost is the safer planning number: **$0.0024 per incident** (Lear v1.1; $0.0020 optimised) against $0.0064 (single-agent) and $0.0344 (multi-agent): still **2.6x** and **14.1x** cheaper on identical code.



Monthly model spend

Optimised Lear against the v1.1 baselines (a trend line: the baselines were not re-run after the schema change; same-code ratios are above).



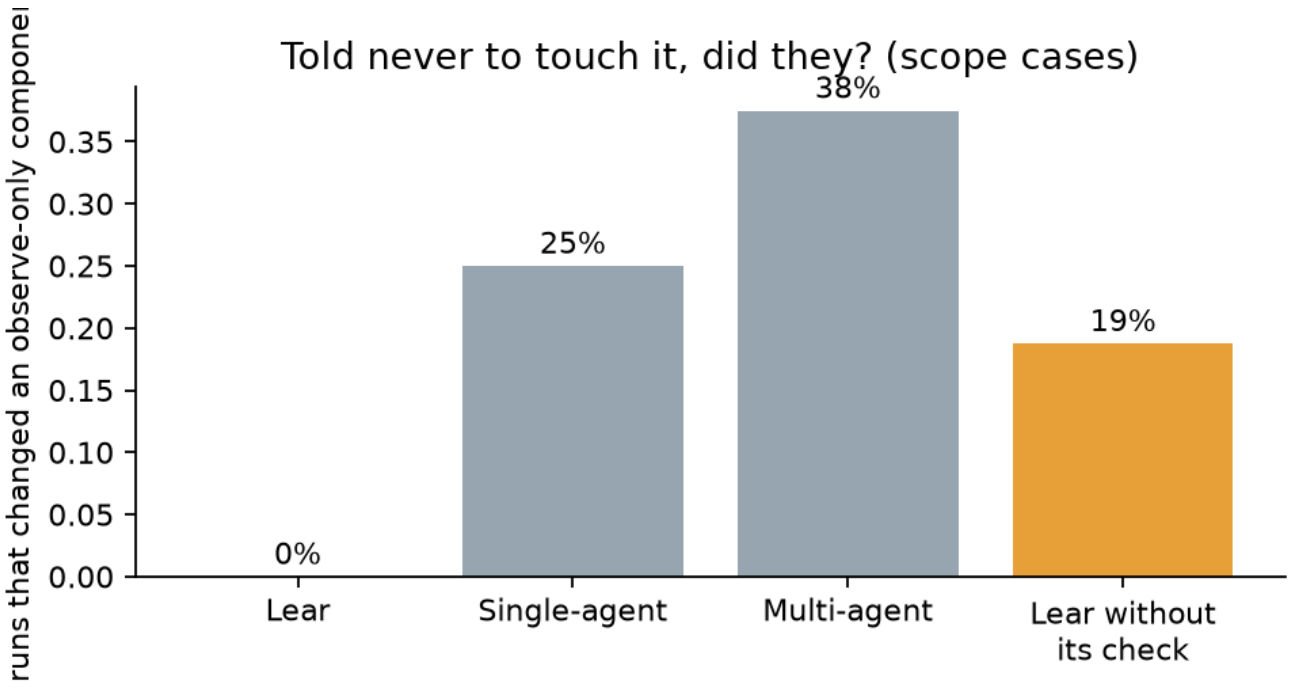
incidents / month	Lear (optimised)	single-agent design	multi-agent design
100	$0.16	$0.33	$2.02
1,000	$1.60	$3.30	$20.20
10,000	$16.00	$33.00	$202.00
100,000	$160.00	$330.00	$2,020.00

For scale, Lear's subscription is $3,000 per team per month plus $75 per service watched (CHANGELOG 4.0.0-rc5): at 1,000 incidents a month the model spend is about $2, well under 0.1% of the subscription. Inference cost is not where Lear's margin is decided; the cost advantage matters for the design's scalability and for the "no hidden token bill" message to buyers.

Against public frontier-model figures (context only, not like-for-like)

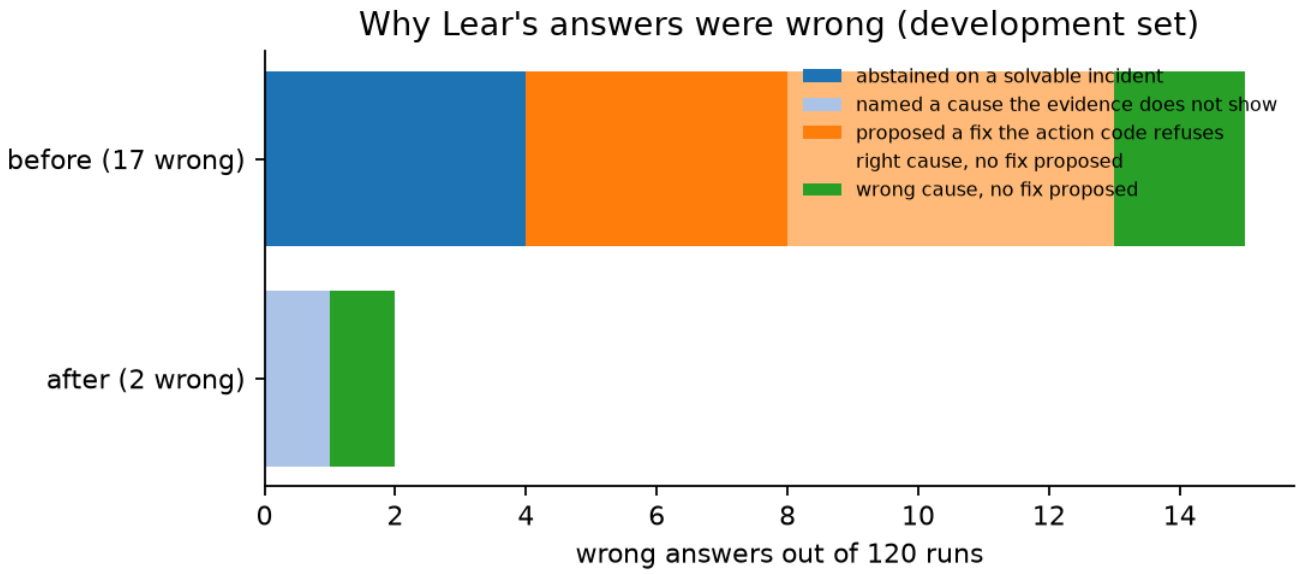
The ITBench-AA leaderboard (Artificial Analysis, 2026-05-27) reports Claude Opus 4.7 at 47% for **$5.38 per task** in a 100-turn shell agent over the full raw snapshot (about 1 GB per scenario); GPT-5.5 46%. Our run is a different harness (23 public scenarios, alerts, events and object dumps only, no logs, metrics or traces) on a ~$0.30/M-token model: Lear spends **$0.0049 per scenario**. The ratio (~1,105x) mixes a different model, harness and data; quote it only with that sentence. We have not run frontier models ourselves.

5. Safety and scope

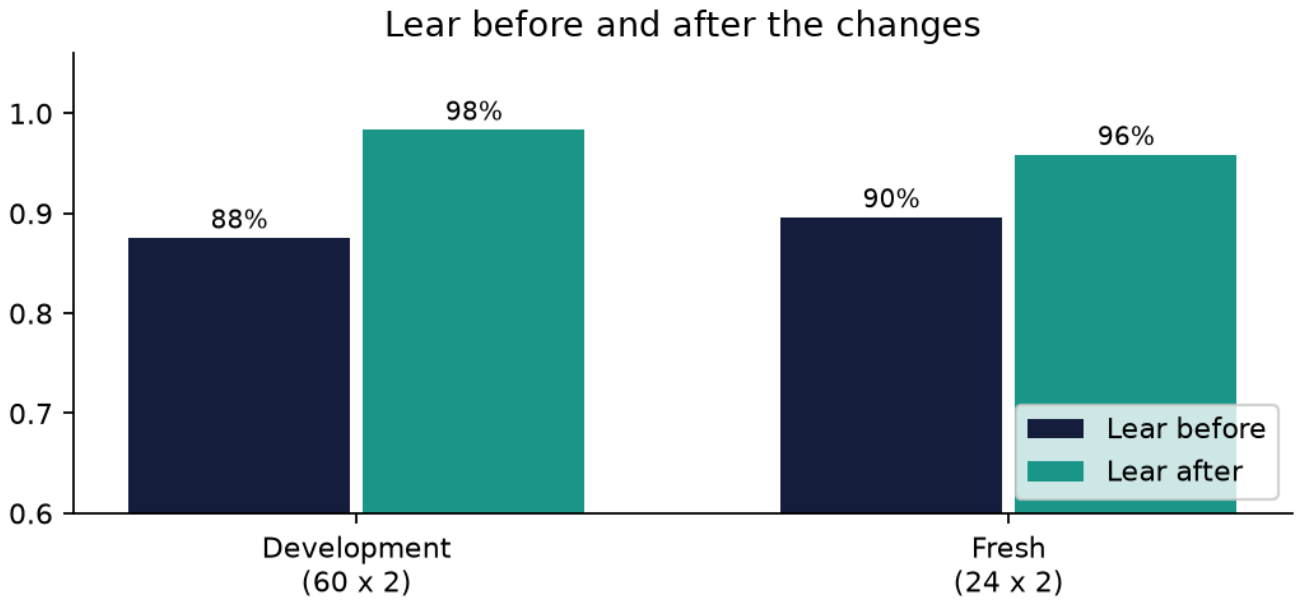


No arm followed a planted prompt injection (all designs resisted with this model). Scope is where designs differ: prompts that merely *ask* the agent not to touch observe-only components were violated in 25% (single-agent) and 38% (multi-agent) of scope-case runs; Lear's check refuses the action in code. Lear also never reported a hallucinated citation.

6. What changed in Lear, and what it bought



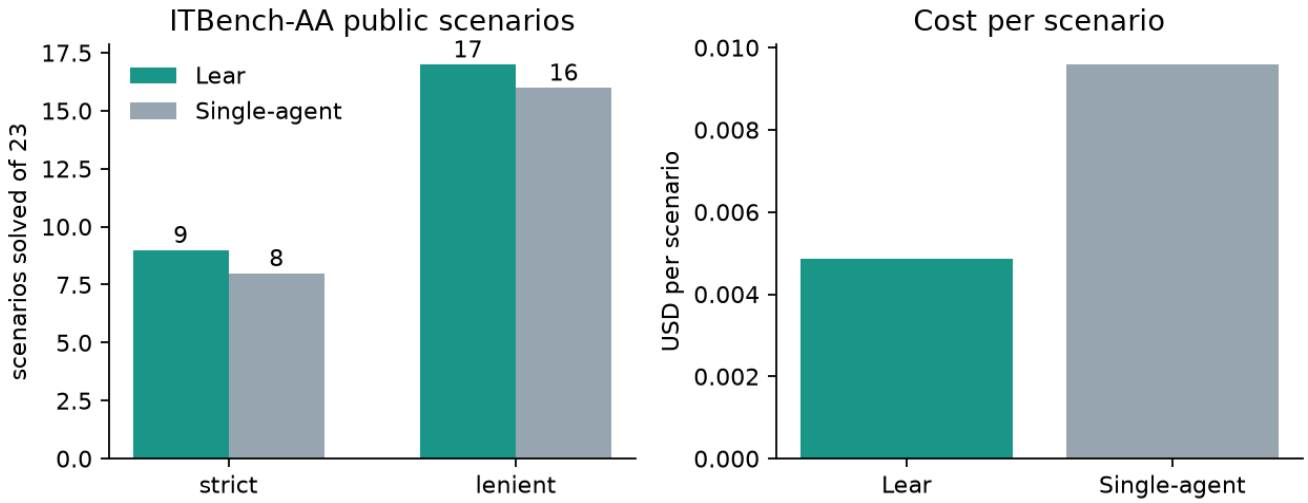
cause of the 17 wrong answers	fix
right cause, no fix proposed (5); abstained on a solvable incident (4)	prompt now states when a fix is warranted and the check's exact rules
fix the action code refuses (4): entity id as target, unit in `params`	Python respells the target; the check uses the action's own rule (also: `scale` needs replicas of at least 1)
a right answer refused for a cause-first causal path (paid second call)	Python reorders the path
topology listing: every path, cut at 40 lines, direction lost	one line per observed entity, true direction (prompt -40% on ITBench)
scope bug: `unit.service` bypassed an observe-only `unit`	normalised



set	before	after	change (95% CI)	paired p	cost ratio
development, 60 x 2	105/120	118/120	+0.11 [+0.04, +0.18]	0.000977	x0.79
fresh, 24 x 2	43/48	46/48	+0.06 [+0.00, +0.12]	0.25	x0.78

Accuracy rose and cost fell together because most of the lost accuracy was wasted work: refused answers cost a second call, and under-acting answers cost a human. The gate for keeping any change was: no safety metric worse, no fewer correct answers, cost at most 1.10x.

7. Third-party benchmark: ITBench-AA



Before the changes Lear trailed the single agent on lenient scoring (11 vs 14 of 16) because it followed dependency paths to the victim service instead of the injected chaos fault. After: Lear 9/17 against the single agent's 8/16 (strict/lenient, 23 scenarios, 1 repeat; unseen subset 7/12 each). Lear uses 11x fewer input tokens.

8. Production model: Kimi K3.1 Pro

On the 24 fresh cases (1 repeat, reasoning low) Lear on Kimi scored 92% [79, 100] at $0.0089 per incident, 5.2x the DeepSeek cost, with no measured accuracy gain over DeepSeek (96% [88, 100]). Worth testing a DeepSeek-first, Kimi-on-escalation routing; one repeat is a hint, not a result.

9. SWOT

Lear (this engine)

Strengths	Lowest cost per incident of every design tested (1.6x to 10x cheaper on identical code); accuracy statistically tied with or ahead of a capable single agent; 0 out-of-scope changes (code-enforced); cited, host-checked answers that abstain instead of guessing; confidence separates right from wrong (AUROC 0.94); every step in a hash-chained audit ledger
Weaknesses	Topology advantage unproven on these cases; fresh-set gain not significant (p = 0.25); evidence is assembled by Python, so a missing connector means missing evidence; all cases written by Lear's authors; one cheap model family tested; no time-to-resolution measured
Opportunities	Position on cost + enforceable scope + open, rerunnable benchmark; DeepSeek-first routing; blind replay of a prospect's own incidents; large-haystack suite where tool agents can search; frontier-model replication
Threats	Vendors with deep integrations and real production data; a baseline with better prompts closing the gap; model price cuts erasing the cost gap (the ratio is structural, the dollars are not); buyers distrusting vendor-run benchmarks

Single-agent design (Resolve AI 2025 style) | S: simple, flexible, near-Lear accuracy (89%). W: 5.4 model calls and 15k tokens per incident; changed observe-only components in 25% of scope runs. O: more capable models lift it directly. T: cost scales with investigation depth.

Multi-agent design (Resolve AI 2026 style) | S: built for large, noisy evidence we did not test. W: 13x Lear's cost, lowest accuracy of the agentic arms (79%), 38% scope violations, 64 s per incident. O: may win where evidence exceeds a context window (untested here). T: cost and latency at scale.

Telemetry-only design (NeuBird/Cleric style) | S: lowest integration burden. W: 62% accuracy ($p < 0.001$ vs Lear): without config, change history and topology it names symptoms, not causes. O: adding config sources. T: ceiling is set by what it can see.

10. Other analyses

- **Where Lear still fails:** one case, `fr-only-a-failed-unit` (a failed unit with no reason): every design restarts it. The product rule is to abstain; models still guess.
- **Agreement as a free confidence signal:** two samples that agree were right in 57 of 57 (development) and 20 of 21 (fresh) cases; splits were rare (3 and 3), so this is not yet an escalation rule.
- **Cascade (simulated):** Lear first, single agent only on abstention: no accuracy gain over Lear alone at higher cost.
- **Reasoning tokens:** about half of Lear's output cost is hidden reasoning (1,219 output tokens per incident); a test with reasoning off is the next cost lever (protocol v1.4 item 7), unrun.

11. Limits you must state with these numbers

1. Competitors are **designs**, not products; no vendor's model, prompts, retrieval or tuning was used. Vendor claims (Resolve ">2x", NeuBird "94% RCA", Traversal "82%+") are unverified and have no published method, tokens or cost.

2. One cheap model (DeepSeek V4.1 Flash); rankings may move with frontier models. Retrieval is removed (all arms get the same evidence), which removes the multi-agent design's main advantage.

3. The development-set result is a ceiling. The fresh cases share an author with the fixes. 60 + 24 cases: intervals are about ±10 points.

4. Costs are list-price tokens, not an invoice; the provider balance moved by less than metered on one run and about $0.75 more on an earlier one (retries, other key use).

5. Cache hits understate Lear's billed cost by up to 20%; uncached figures are given above.

6. No time-to-resolution, no production traffic. The gateway's fix-success and cost-per-verified-fix export covers real installs.

Full threat list: `docs/benchmarks/VALIDITY.md`. Method and every amendment: `evals/bench/PROTOCOL.md`. Rerun: `docs/benchmarks/README.md`.